

Network Diagnostics Reference Manual

A. HTTP Header Inspection

```
curl -I www.gemini.com
```

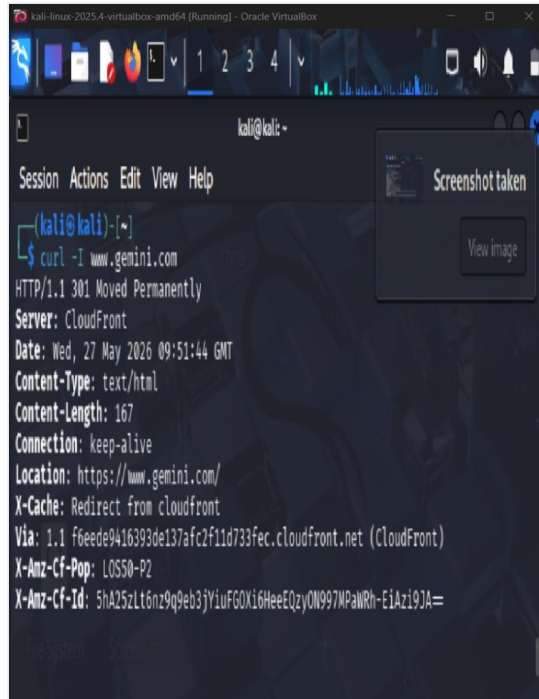
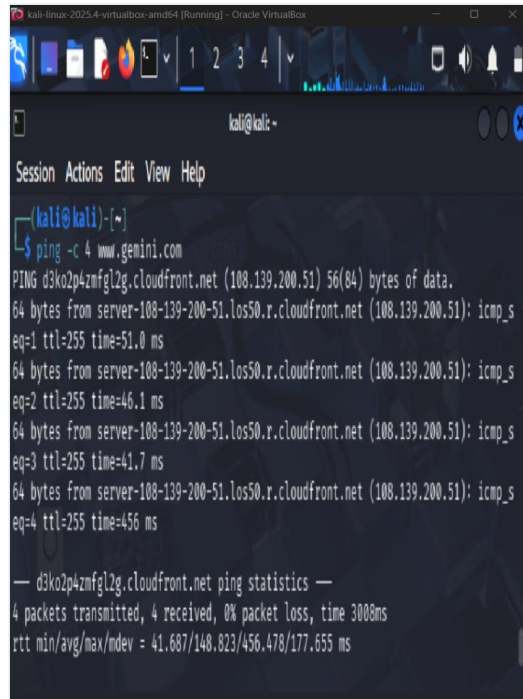


Figure 1: Capturing edge load-balancer signatures and CloudFront cache hits.

B. ICMP Echo Testing

```
ping -c 4 www.gemini.com
```



```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali) ~  
$ ping -c 4 www.gemini.com  
PING d3ko2p4zmfgl2g.cloudfront.net (108.139.200.51) 56(84) bytes of data.  
64 bytes from server-108-139-200-51.los50.r.cloudfront.net (108.139.200.51): icmp_s  
eq=1 ttl=255 time=51.0 ms  
64 bytes from server-108-139-200-51.los50.r.cloudfront.net (108.139.200.51): icmp_s  
eq=2 ttl=255 time=46.1 ms  
64 bytes from server-108-139-200-51.los50.r.cloudfront.net (108.139.200.51): icmp_s  
eq=3 ttl=255 time=41.7 ms  
64 bytes from server-108-139-200-51.los50.r.cloudfront.net (108.139.200.51): icmp_s  
eq=4 ttl=255 time=456 ms  
  
— d3ko2p4zmfgl2g.cloudfront.net ping statistics —  
4 packets transmitted, 4 received, 0% packet loss, time 3000ms  
rtt min/avg/max/mdev = 41.687/148.823/456.478/177.655 ms
```

Figure 2: Verifying latency and 0% packet loss.

C. Network Path Traceroute

```
traceroute www.gemini.com
```



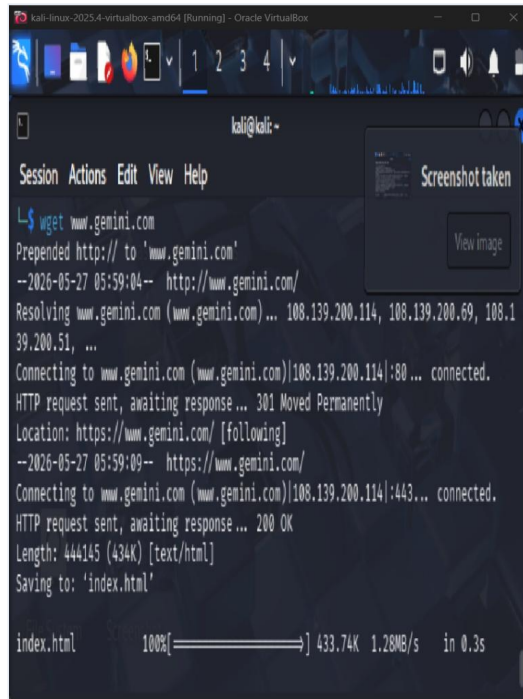
The screenshot shows a terminal window titled 'kali-linux-2023.4-virtualbox-amd64 [Running] - Oracle VM VirtualBox'. The terminal prompt is 'kali@kali: ~'. The user has entered the command 'traceroute www.gemini.com'. The output shows the path from the local host to the destination IP 108.139.200.69. The first hop is 10.0.2.2 (10.0.2.2) with three round-trip times: 4.492 ms, 6.086 ms, and 4.638 ms. Hops 2 through 11 are marked with three asterisks (***) indicating successful but unresponsive hops.

```
kali@kali: ~  
Session Actions Edit View Help  
--(kali@kali)-[~]  
-$ traceroute www.gemini.com  
traceroute to www.gemini.com (108.139.200.69), 30 hops max, 60 byte packets  
1 10.0.2.2 (10.0.2.2) 4.492 ms 6.086 ms 4.638 ms  
2 ***  
3 ***  
4 ***  
5 ***  
6 ***  
7 ***  
8 ***  
9 ***  
10 ***  
11 ***
```

Figure 3: Diagnosing intermediate physical network routers.

D. Content Procurement

```
wget www.gemini.com
```



```
kali@kali:~$ wget www.gemini.com
Prepended http:// to 'www.gemini.com'
--2026-05-27 05:59:04-- http://www.gemini.com/
Resolving www.gemini.com (www.gemini.com)... 108.139.200.114, 108.139.200.69, 108.139.200.51, ...
Connecting to www.gemini.com (www.gemini.com)|108.139.200.114|:80... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://www.gemini.com/ [following]
--2026-05-27 05:59:09-- https://www.gemini.com/
Connecting to www.gemini.com (www.gemini.com)|108.139.200.114|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 444145 (434K) [text/html]
Saving to: 'index.html'

index.html  100%[=====] 433.74K  1.28MB/s  in 0.3s
```

Figure 4: Automated secure HTTPS document request.